

Train / Validation / Test Split — Group Discussion

For each of the five datasets below:

- 1. Decide **what one "unit" of the dataset is**.
- 2. Based on that unit, discuss with your group **how you would split the data into train / validation / test, and why**.

Dataset A: Monthly Electricity Demand

An energy utility wants to forecast monthly electricity demand for a metropolitan region. One row per month, from Jan 2005 to Dec 2024. The company wants to know how well the model predicts demand for 2025 and beyond. Demand shows a clear upward trend and strong seasonality.

month	avg_temp(°C)	num_holidays	industrial_index	population	demand_MWh
2005-01	2.1	2	98.4	2,410,000	1,842,000
2005-02	3.5	1	97.9	2,411,500	1,760,300
2005-03	8.9	1	99.1	2,413,000	1,690,100
2005-04	14.2	2	100.3	2,414,800	1,588,400
...	...	...	...	...	...
2024-09	23.6	1	130.5	2,978,900	2,463,200
2024-10	16.4	2	130.9	2,979,700	2,401,700
2024-11	9.8	1	131.2	2,980,400	2,551,900
2024-12	1.3	3	129.7	2,982,100	2,742,600

Dataset B: Activity Recognition from Wearables

A company recognizes physical activities (walking, running, sitting, cycling) from wearable-sensor signals. Data comes from **150 volunteers**, each wearing the device for a week, producing many time-windowed samples per person. People differ a lot: gait, body size, sensor placement, and baseline heart rate vary strongly. The product will be sold to **new customers who never contributed data**.

window_id	subject_id	acc_x_mean	acc_x_std	gyro_z_std	heart_rate	activity
W_000031	S_003	0.12	0.88	1.42	78	walking
W_000032	S_003	0.15	0.91	1.55	81	walking
W_000033	S_003	0.14	0.86	1.49	80	walking
...	...	...	...	...	...	...
W_004410	S_003	-0.02	0.31	0.44	66	sitting
W_004411	S_003	-0.01	0.29	0.41	65	sitting
...	...	...	...	...	...	...
W_009120	S_087	0.44	1.96	3.81	142	running
W_009121	S_087	0.47	2.01	3.90	145	running
W_009122	S_087	0.45	1.98	3.86	144	running
...	...	...	...	...	...	...
W_015803	S_142	0.09	0.72	2.14	96	cycling
W_015804	S_142	0.11	0.75	2.20	98	cycling
...	...	...	...	...	...	...

Dataset C: Email Promotion Response

A retail company wants to predict whether a customer will respond to a promotional email. Each row is one customer. Customers were sampled independently from the national database, there is no time stamp, and no customer appears more than once. The target is roughly balanced (about 45% / 55%). The model must predict response for the general customer population.

customer_id	age	income(USD)	past_purchases	avg_basket(USD)	email_open_rate	responded
C_00412	34	52,000	7	68.40	0.42	1
C_18830	61	88,500	23	142.10	0.71	1
C_07751	27	31,200	1	24.90	0.08	0
C_29104	45	67,900	12	95.30	0.55	0
C_11627	52	120,000	40	210.75	0.83	1
C_03390	38	44,600	5	51.20	0.30	0
C_25518	29	58,300	9	73.85	0.61	1
C_16204	47	72,100	15	118.40	0.24	0
...	...	...	...	...	...	...

Dataset D: Citation Network Link Prediction

Researchers have a **citation network** of academic papers (e.g. the Cora / Citeseer / arXiv style benchmark). The data is a **graph**: each node is a paper (with features such as its topic and a bag-of-words summary of its abstract), and an edge means **one paper cites another**. The task is **link prediction** — given the papers and the citations we can see, predict which other paper pairs are linked by a citation.

The data comes as two tables: a node table and an edge (citation) table.

Node table — one row per paper.

paper_id	subject	year	word_present_0	word_present_1	...	word_present_1432
P_0001	Neural_Networks	2011	0	1	...	0
P_0002	Reinforcement_Learning	2013	1	0	...	1
P_0003	Theory	2009	0	0	...	0
P_0004	Neural_Networks	2015	1	1	...	0
P_0005	Probabilistic_Methods	2012	0	1	...	1
...	...	...	...	...	...	...

Edge table — one row per citation link.

citing_paper	cited_paper
P_0001	P_0003
P_0004	P_0001
P_0002	P_0005
P_0004	P_0003
...	...

Dataset E: Credit-Card Fraud Detection

A bank wants to detect fraudulent credit-card transactions. Each row is one transaction. Out of 2,000,000 transactions, only about **1,200 (0.06%)** are fraud. The team needs a model that reliably catches the rare fraud cases.

txn_id	amount(\$)	merchant_cat	time_since_last(s)	location	is_fraud
...	...	...	...	...	...
T_1000001	42.10	grocery	3,600	San Francisco	0
T_1000002	12.99	coffee	780	Dixon	0
T_1000003	1,890.00	electronics	12	Vacaville	1
T_1000004	7.50	transit	1,420	Winters	0
T_1000005	63.20	restaurant	5,900	Sacramento	0
T_1000006	129.00	clothing	2,300	Sacramento	0
T_1000007	2,450.00	gift_card	5	Vacaville	0
T_1000008	18.75	pharmacy	940	Davis	0
T_1000009	9.99	streaming	86,400	San Francisco	0
...	...	...	...	...	...